

Bias Collapse in Shallow ReLU Networks

Numerical Evidence for Open Problems 4.1, 4.2, and 4.3

NSF RTG

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Motivation

The overparameterization puzzle:

- ▶ Modern networks have far more parameters than the problem demands
- ▶ Classical learning theory predicts overfitting; practice shows robust generalization
- ▶ Why does the optimizer self-simplify?

Bias collapse: a concrete mechanism

- ▶ Gradient flow automatically eliminates redundant neurons, with no regularization required
- ▶ $m = 1500$ neurons $\rightarrow$ effective complexity of $k = 7$ neurons
- ▶ The number of surviving *bias clusters* equals the number of *inflection points* of the target function

This work

Rigorous numerical investigation of three open problems:

1. **OP 4.1:** Does $C(m, f^*) \rightarrow k$ as $m \rightarrow \infty$?
2. **OP 4.2:** Does collapse generalize to $d \geq 2$ dimensions?
3. **OP 4.3:** Is there a provable L^2 pruning bound after collapse?

72 runs, 9 targets, m up to 5000.

Model and Gradient Flow

1D shallow ReLU network:

$$f(x) = \sum_{j=1}^m a_j \sigma(x - b_j), \quad \sigma = \max(0, \cdot), \quad x \in [-1, 1]$$

Each neuron places a *kink* at bias b_j with slope change a_j .

Gradient flow on MSE $\mathcal{L} = \frac{1}{2} \int_{-1}^1 (f - f^*)^2 dx$:

$$\begin{aligned} \dot{a}_j &= - \int_{-1}^1 (f - f^*) \sigma(x - b_j) dx \\ \dot{b}_j &= a_j \int_{b_j}^1 (f - f^*) dx \end{aligned}$$

Solved via `scipy` RK45, $T = 500$, $\text{rtol} = 10^{-4}$.

Stationarity criterion: $\max_j |\dot{a}_j|, |\dot{b}_j| < 0.01$.

Key measured quantity:

$$C(m, f^*) = \#\{\text{bias clusters}\}$$

at stationarity; gap threshold 0.02.

Fixed-point condition

Each *active* neuron ($a_j \neq 0$) satisfies $R_j := \int_{b_j}^1 (f - f^*) dx \approx 0$ at stationarity. Verified numerically at every converged run; connects to the theory of OP 4.1.

The Three Open Problems

OP 4.1: Cluster Count

$$\lim_{m \rightarrow \infty} C(m, f^*) = k$$

Does the number of bias clusters converge to the number of sign-changing inflection points of f^* ?

OP 4.2: Higher Dimensions

For $f^* : [-1, 1]^2 \rightarrow \mathbb{R}$, does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

OP 4.3: Pruning Bound

After collapse, is there a provable L^2 error bound when pruning to k neurons?

$$\|\tilde{f} - f\|_{L^2} \leq \delta \cdot \sum_j |a_j|$$

Each open problem is introduced in detail immediately before its results.

Open Problem 4.1: Cluster Count Conjecture

Conjecture

$$\lim_{m \rightarrow \infty} C(m, f^*) = k$$

where $k := \#$ sign-changing inflection points of f^* in $(-1, 1)$

$C(m, f^*)$ = number of distinct bias clusters at stationarity.

k = number of sign-changing inflection points of the **target** f^* .

f^* = the target function (e.g. $\sin(3\pi x)$).

Sub-problems we address:

- ▶ Can we prove $C(m, f^*) \leq k$ independently of m ?
- ▶ Which neurons survive collapse to become cluster representatives?

Experimental Design

Nine target functions:

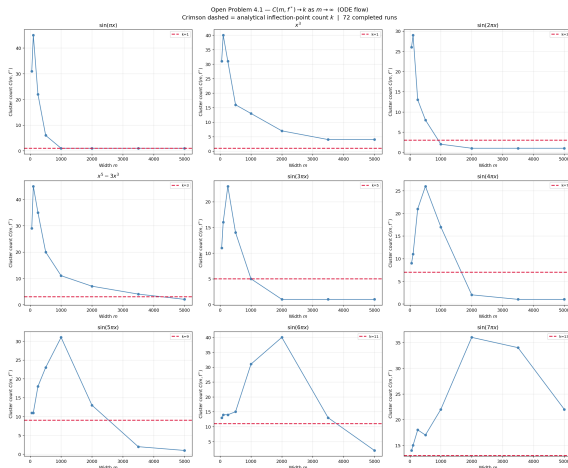
$f^*(x)$	k	type
$\sin(\pi x)$	1	trig
x^3	1	poly
$\sin(2\pi x)$	3	trig
$x^5 - 3x^3$	3	poly
$\sin(3\pi x)$	5	trig
$\sin(4\pi x)$	7	trig
$\sin(5\pi x)$	9	trig
$\sin(6\pi x)$	11	trig
$\sin(7\pi x)$	13	trig

Trig: $k = 2n - 1$. Poly: irregular inflection spacing.

Main sweep:

- ▶ $m \in \{50, 100, 250, 500, 1000, 2000, 3500, 5000\}$: number of neurons in the network
- ▶ $T = 500$: time horizon for ODE integration. SciPy's RK45 solver, a standard adaptive step-size integrator, numerically solves the gradient flow equations from $t = 0$ to $t = T$, automatically adjusting step size to maintain accuracy.
- ▶ **72 completed runs** across 9 targets

OP 4.1: $C(m, f^*)$ Across All 9 Targets



$C(m, f^*)$ vs. m at $T = 500$ (72 runs, 9 targets). **Crimson dashed line = k .** All panels show rise-then-fall. Low- k trig targets reach $C = k$; $\sin(7\pi x)$ ($k = 13$) reaches $C = 13$ at $m = 5000$ for the first time (stationarity unconfirmed).

OP 4.1: The Rise-Then-Fall Dynamic

Universal pattern across all 9 targets:

1. **Rise:** Small m cannot populate all k inflection attractors simultaneously. C grows above k .
2. **Peak:** C reaches a maximum scaling with k (up to $C = 40$ for $k = 11$).
3. **Fall:** With sufficient redundancy, collapse proceeds and C decreases toward k .

Key observation

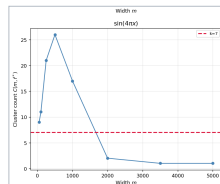
The peak of $C(m)$ shifts to **larger** m as k grows. High- k targets require much wider networks before the fall phase begins.

Selected $C(m)$ at $T = 500$:

Target	k	$m=1000$	$m=2000$	$m=5000$
$\sin(\pi x)$	1	1	1	1
$\sin(3\pi x)$	5	5	†1	†1
$\sin(4\pi x)$	7	17	2	†1
$\sin(7\pi x)$	13	22	36	‡13
$x^5 - 3x^3$	3	11	7	2
x^3	1	13	7	4

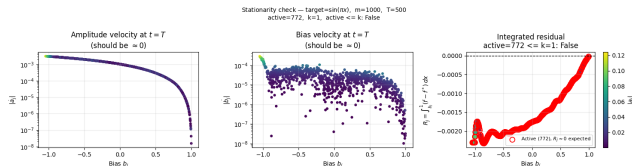
† Cluster structure absent or too dense for fixed threshold to detect.

‡ $C = k$ but not fully stationary ($\max |\dot{a}_j| = 0.034$).



↪ $\sin(4\pi x)$, $k = 7$: rise-then-fall

OP 4.1: Stationarity and Fixed-Point Verification



$\sin(\pi x)$, $m = 1000$, $T = 500$

Final (a_j, b_j) substituted back into the ODEs.

Left: $|\dot{a}_j| \approx 10^{-3} - 10^{-5}$

Center: $|\dot{b}_j| \approx 10^{-4} - 10^{-5}$

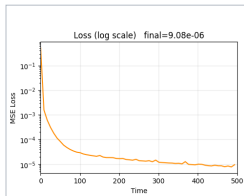
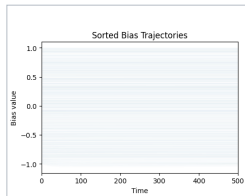
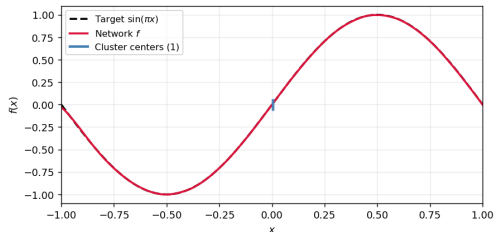
Right: $R_j \approx 0$ for active neurons (red), confirming the theoretical fixed-point condition numerically.

Discrete GD comparison

Discrete GD does *not* produce $C \rightarrow k$ for any target except $\sin(\pi x)$. The conjecture is specific to continuous ODE gradient flow dynamics.

OP 4.1 Confirmation: $\sin(\pi x)$, $k = 1$

target = $\sin(\pi x)$, $m = 1000$, $T = 500$
clusters = 1, $k = 1$, $C = k$: True



$\sin(\pi x)$, $m = 1000$, $k = 1$

Top: Clean final fit vs. target. Blue ticks = single cluster center.

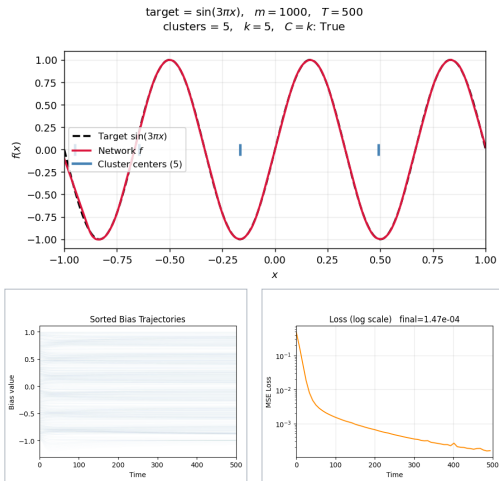
Bottom: Bias trajectories (all 1000 collapse to **one location**) and MSE loss on log scale.

Strongest confirmation

$C = 1 = k$ for *all* $m \geq 1000$ through $m = 5000$, fully stationary ($\max |\dot{a}_j| < 3.6 \times 10^{-3}$).

Only target with $C = k$ confirmed robustly across the full m sweep.

OP 4.1 Confirmation: $\sin(3\pi x)$, $k = 5$

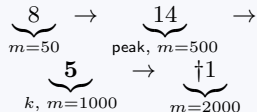


$\sin(3\pi x)$, $m = 1000$, $k = 5$

Top: Clean final fit. Five blue ticks mark the **5 bias clusters**, one per inflection point.

Bottom: Bias trajectories and MSE loss on log scale.

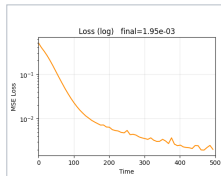
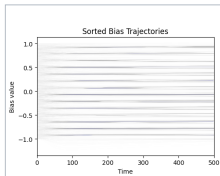
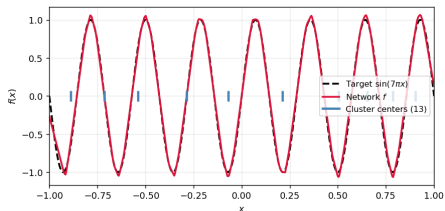
Non-monotone path to k



Rises above k , reaches $k = 5$, then $\dagger$ cluster structure dissolves at larger m .

OP 4.1: $\sin(7\pi x)$, $k = 13$ (Most Complex Target)

target = $\sin(7\pi x)$, $m = 5000$, $T = 500$
clusters = 13, $k = 13$, $C = k$: True



$\sin(7\pi x)$, $m = 5000$, $k = 13$

Top/Bottom: At $T = 500$, the fit shows 13 clusters aligning with inflection points, but the network has not settled.

C at $m = 5000$

$T = 500$: $C = 13 = k$, $\max |\dot{a}_j| = 0.034$

$T = 1000$: $C = 22$, $\max |\dot{a}_j| = 0.077$

$C = 13$ was **transient**: the run has not stabilized at $m = 5000$, $T = 500$.

What is needed

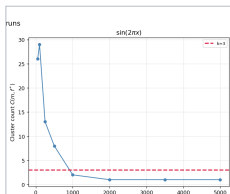
Confirmed $C \rightarrow k$ for $k = 13$ likely requires $m \gg 5000$. Larger m provides the redundancy for stable collapse; longer T alone is insufficient if m is too small.

OP 4.1: The Below- k Phenomenon

Three cases at verified stationary states with $C < k$. Re-run with 2 additional seeds each:

Target	k	m	seed 42	seed 5	seed 25
$\sin(2\pi x)$	3	1000	2	1	1
$\sin(4\pi x)$	7	2000	2	3	5
$x^5 - 3x^3$	3	5000	2	2	$3 = k$

C varies across seeds in all three cases. No single $C < k$ value is consistent.



$\hookrightarrow \sin(2\pi x), k = 3: C$ stays below k

Interpretation

C varies across seeds $\Rightarrow C < k$ states are **initialization-dependent local minima**, not universal fixed points.

Gradient flow has multiple basins of attraction. The k -cluster basin exists but is not the only one.

$x^5 - 3x^3$ seed 25: $C = k = 3$

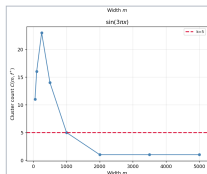
First time $x^5 - 3x^3$ reaches $C = k$. Shows the k -cluster attractor exists for polynomial targets, though harder to find.

Instability Near k : Results

Question: Are states with $C \neq k$ unstable? Does gradient flow push them toward $C = k$?

22 qualifying runs: 8 ($C = k$), 8 ($C > k$), 6 ($C < k$).
 $T_{\text{perturb}} = 1000$.

Category	Status	Finding
$C > k$ states	8/8 done	No change. C stays fixed; ODE fixed points do not dissolve.
$C < k$ states	12/12 done	No change. Injected neuron amplitude stays at 0.01 and cannot grow.
$C = k$ states	16/16 done	All returned to k. Exact- k states are stable attractors.



$\hookrightarrow \sin(3\pi x)$, $k = 5$: $C = k$ state is stable

$C > k$: stable fixed points

Continuing ODE for $T = 1000$: C does **not** decrease.
 $C > k$ states are *stable* fixed points, not metastable.

$C < k$: injection-resistant

Injected neurons gain **zero amplitude**. ODE provides no restoring force toward k . Supports: $C < k$ states are true fixed points.

Takeaway

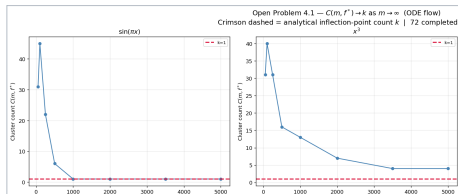
Both $C > k$ and $C < k$ states are stable fixed points. A proof of $C \rightarrow k$ cannot argue that other states dissolve on their own; it must explain why random initializations land at $C = k$.

OP 4.1: Polynomial vs. Trigonometric Targets

Same k , very different convergence speed:

Target	k	$m=1000$	$m=2000$	$m=5000$
$\sin(\pi x)$	1	1	1	1
x^3	1	13	7	4
$\sin(2\pi x)$	3	2	†1	†1
$x^5 - 3x^3$	3	11	7	2

x^3 reaches only $C = 4$ at $m = 5000$, despite $k = 1$.



↔ $\sin(\pi x)$ (left) vs x^3 (right), same $k = 1$

Why inflection geometry matters

For $\sin(n\pi x)$, the k inflection points are *evenly spaced* across $[-1, 1]$, forming strong, symmetric attractors that neurons organize around early in training.

Polynomial targets have *irregular* spacing: weaker, asymmetric attractors. Neurons take far longer to find them.

Implication: a proof of OP 4.1 must account for attractor geometry, not just count.

OP 4.1: Key Findings

What was confirmed

- ▶ $C \rightarrow k$ confirmed for $\sin(\pi x)$ ($k = 1$), all $m \geq 1000$ (strongest case).
- ▶ $\sin(7\pi x)$: $C = 13$ at $m = 5000$, $T = 500$ was transient ($C = 22$ at $T = 1000$). Confirmed $C \rightarrow k$ requires $m \gg 5000$.
- ▶ $C = k$ states are stable attractors: all 16 perturbation runs returned to $C = k$.

What the data reveals about OP 4.1

- ▶ $C < k$ local minima exist for 3 targets at genuinely stationary states.
- ▶ $C > k$ and $C < k$ states are both stable fixed points; a proof of $C \rightarrow k$ must rely on **initialization geometry**, not instability.
- ▶ Polynomial targets converge far more slowly due to irregular attractor geometry.

Open Problem 4.2: Higher-Dimensional Collapse

Question

For a structured target $f^* : [-1, 1]^2 \rightarrow \mathbb{R}$, does gradient flow concentrate the neuron measure onto a lower-dimensional set in hyperplane parameter space?

2D network:

$$f(x) = \sum_{j=1}^m a_j \sigma(w_j^\top x + b_j), \quad x \in [-1, 1]^2$$

Each neuron defines an oriented hyperplane (line in 2D). Scale-invariant: $\theta_j = \angle(w_j)$, $\beta_j = b_j / \|w_j\|$.

Collapse in 2D: the (θ_j, β_j) cloud concentrates onto a low-dimensional subset.

Diagnostic

Angular entropy $H(\{\theta_j\})$: near 0 if collapsed; near $\log(36) \approx 3.58$ if uniform. Effective neurons $N_{\text{eff}} = e^H$:
 $N_{\text{eff}}=1$ full collapse; $N_{\text{eff}} \approx 36$ uniform.

Width sweep

$m \in \{64, 256, 512, 1024\}$, 6000 epochs, Adam.

Three targets studied

Ridge, Separable, and Radial.
See next slide for expected vs. observed behavior.

Open Problem 4.2: Target Functions

Ridge

$$\sin(2\pi u^\top x) + 0.5 \sin(4\pi u^\top x)$$

Expected (small width):
concentration near $\theta \approx 53^\circ$

Observed:
concentration weakens as width increases

Separable

$$\sin(2\pi x_1) + 0.3 \sin(2\pi x_2)$$

Expected (small width):
two dominant direction families

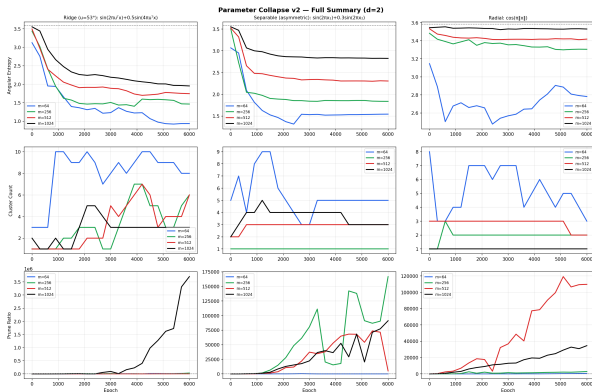
Observed:
directional structure becomes diffuse at large width

Radial (reference)

$$\cos(\pi \|x\|)$$

Reference case:
approximately isotropic orientation distribution

OP 4.2: Orientation Distributions



Entropy H / Eff. neurons $N_{\text{eff}} = e^H$ at epoch 6000:

Target	$m=64$	$m=256$	$m=512$	$m=1024$
Ridge	0.94 / 3	1.46 / 4	1.74 / 6	1.95 / 7
Separable	1.55 / 5	1.84 / 6	2.31 / 10	2.83 / 17
Radial	2.78 / 16	3.31 / 27	3.42 / 31	3.53 / 34

$N_{\text{eff}}=1$: full collapse; $N_{\text{eff}} \approx 36$: uniform. Max $H = \log(36) \approx 3.58$

Open question

Why does directional concentration weaken as width grows? Observed collapse appears strongest in moderately overparameterized networks. The mechanism behind this trend remains unclear.

OP 4.2: Key Findings

What the data shows

- ▶ At small width, directional concentration reflects target geometry:
ridge $\rightarrow$ one direction family;
separable $\rightarrow$ two families;
radial $\rightarrow$ broadly distributed.
- ▶ Concentration decreases steadily as width grows.
- ▶ Networks increasingly fit targets using many distinct directions.

Effective directional complexity

$N_{\text{eff}} = e^H$ measures how many distinct directions the network actively uses.

- ▶ N_{eff} increases monotonically with width.
- ▶ Wider networks rely less on directional collapse.
- ▶ Radial $m=1024$: $N_{\text{eff}} \approx 34$ of 36 (nearly uniform).

Open Problem 4.3: Pruning Bound

Bound (from the slides)

$$\|\tilde{f} - f\|_{L^2} \leq \delta \cdot \sum_{j=1}^m |a_j|$$

Setup: After gradient flow, bias clusters have formed. *Prune* by replacing each cluster with a single representative neuron, giving $\tilde{f}$ with only k neurons.

δ = maximum intra-cluster diameter (how spread out the biases are within each cluster).

$\sum |a_j|$ = total weight mass of the full m -neuron network.

Why it matters: Provides a formal L^2 certificate that collapsing from m neurons to k neurons is safe, bridging the collapse phenomenon and approximation theory.

Why the bound holds

Within a cluster, all b_j differ by at most δ , so $\sigma(x - b_j) \approx \sigma(x - b_{\text{center}})$ up to an error $\leq \delta$. Summing over m neurons, errors accumulate at most as $\delta \cdot \sum |a_j|$.

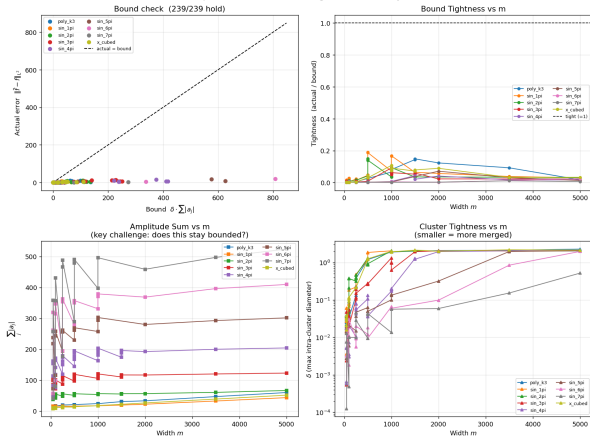
The challenge

For the bound to be *non-vacuous* (goes to 0 as networks grow), we need $\delta \cdot \sum |a_j| \rightarrow 0$.

Our data shows $\sum |a_j| \sim \sqrt{m}$. So we need $\delta \rightarrow 0$ fast enough, which must come from longer training T , not wider networks m .

OP 4.3: Pruning Bound Verification

Open Problem 4.3 — Pruning Bound Summary



$$\text{Bound: } \|\tilde{f} - f\|_{L_2} \leq \delta \cdot \sum_j |a_j|$$

72 / 72 runs: bound holds

Tightness (actual / bound): 0.0002 to 0.148. Substantial slack in all cases.

Key obstacle:

- ▶ $\sum_j |a_j| \sim \sqrt{m}$ empirically
- ▶ δ does not shrink with m at $T = 500$
- ▶ Bound loosens $\sim \sqrt{m}$ while actual error plateaus

Path to non-vacuousness

$\delta \rightarrow 0$ requires $T \rightarrow \infty$, *not* $m \rightarrow \infty$.
Study $\delta(T)$ at fixed m to make the bound formally useful.

Lottery Ticket: Which Neurons Survive?

Setup: At $t = 0$, amplitudes $a_j \sim \mathcal{N}(0, 0.01)$ are random. Only the bias placement b_j is structured. Neurons near an inflection point are *geometrically lucky*.

Five conditions per (target, m): full, geometric, bias_only, amp_only, random_k
52 runs; 4 targets, $m \in \{500, 1000, 1500\}$.

Claim 2 (performance): FALSE

Lucky k neurons alone achieve $100\times$ – $50,000\times$ higher loss than the full network. k kinks cannot approximate a smooth target; full redundancy is necessary.

Claim 1 (survival): bias position is key

geometric $\approx$ **bias_only** in every case (loss diff < 0.002). Initial amplitudes carry *no information*.

amp_only is the worst condition: good amplitudes with random biases performs worse than good biases with random amplitudes.

Conclusion:

Bias position at $t = 0$ is the sole informative quantity for collapse. The “winning ticket” is a good bias placement, not a good amplitude.

Summary of Evidence

Problem	Main finding	Status
OP 4.1: $C \rightarrow k$	Confirmed for $\sin(\pi x)$ ($k = 1$), all $m \geq 1000$. $\sin(7\pi x)$: $C = 13$ at $m = 5000$, $T = 500$ was transient ($C = 22$ at $T = 1000$); $m \gg 5000$ needed. Below- k : init-dependent local minima (multiple seeds).	Supported for $k = 1$; open for $k \geq 3$
Instability near k	$C > k$: stable fixed points (8/8, no dissolution). $C < k$: injection-resistant (12/12, zero amplitude growth). $C = k$: all 16 returned to k (stable attractors).	Complete
OP 4.2: 2D collapse	Directional concentration tracks target structure at small width. Decreases consistently with width ($m \in \{64, 256, 512, 1024\}$). Effective directional complexity (N_{eff}) increases with m .	Partial
OP 4.3: Pruning bound	72 / 72 hold. $\sum a_j \sim \sqrt{m}$. Non-vacuousness requires $\delta \rightarrow 0$ as $T \rightarrow \infty$.	Verified numerically ; open theoretically
OP 4.1.4: Lottery ticket	Bias position at $t = 0$ is sole informative quantity. Amplitude values irrelevant. Performance claim false.	Resolved

Open Questions and Next Steps

OP 4.1 (open):

- ▶ A proof of $C \rightarrow k$ must account for initialization geometry. Below- k local minima exist, so the k -cluster basin must be shown to dominate at large m
- ▶ **4.1.1:** $C > k$ states are *stable* fixed points, so instability alone cannot drive $C \rightarrow k$
- ▶ $\sin(7\pi x)$: Re-run at $T = 1000$ to confirm stationarity of $C = 13$

OP 4.2 (open):

- ▶ Why does directional concentration weaken as width grows? The mechanism behind this trend remains unclear.

OP 4.3, path to a formal proof:

- ▶ Study $\delta(T)$ at fixed m : does $\delta \rightarrow 0$ as $T \rightarrow \infty$?
- ▶ Exponent α in $\sum_j |a_j| = O(m^\alpha)$; data suggests $\alpha \approx 1/2$
- ▶ If $\delta = O(T^{-\gamma})$ the bound becomes non-vacuous as $T \rightarrow \infty$

Immediate priorities

- (1) $\sin(7\pi x)$ at $T = 1000$ (confirm $C = 13$)
- (2) $\delta(T)$ trajectory study
- (3) Theory for capacity-constrained collapse

Conclusion

What we showed:

- ▶ $C \rightarrow k$ confirmed for $\sin(\pi x)$ ($k = 1$), $m \geq 1000$. First high- k evidence: $\sin(7\pi x)$ reaches $C = 13$ at $m = 5000$.
- ▶ **Rise-then-fall** is universal: peak scales with k ; inflection geometry sets the convergence rate.
- ▶ **Below- k** : 3 targets have $C < k$ at genuinely stationary states, an open challenge for OP 4.1.
- ▶ **Instability**: Above- k states are stable fixed points. Proof of $C \rightarrow k$ must use initialization geometry.
- ▶ **OP 4.2**: Effective directional complexity grows with width; wider networks rely less on orientation collapse than narrower ones.
- ▶ **OP 4.3**: Bound holds 72/72; $\delta \rightarrow 0$ with T is the path to a formal proof.
- ▶ **Lottery ticket**: bias position at $t = 0$ is the sole informative quantity.

Broader implication

Gradient flow is not only an optimizer; it is a **compression algorithm** that discovers the minimum-complexity representation of the target's intrinsic structure.

Inflection points of f^* act as **fixed-point attractors**: neurons near one survive; the rest dissolve.

A rigorous proof would explain generalization in overparameterized networks.

Thank you

Repository & documentation github.com/calebabr/NSF-RTG-Project4 full scripts, data, and result tables in README.md